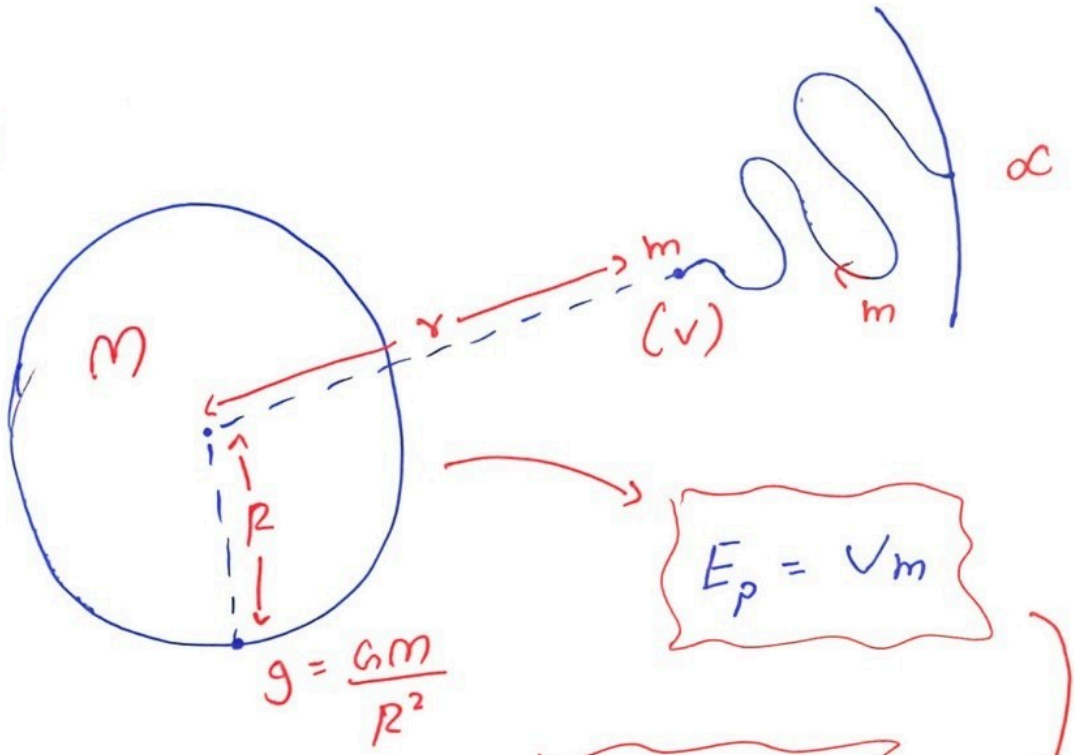
Gravitational field

⟹ Gravitational Potential energy

①

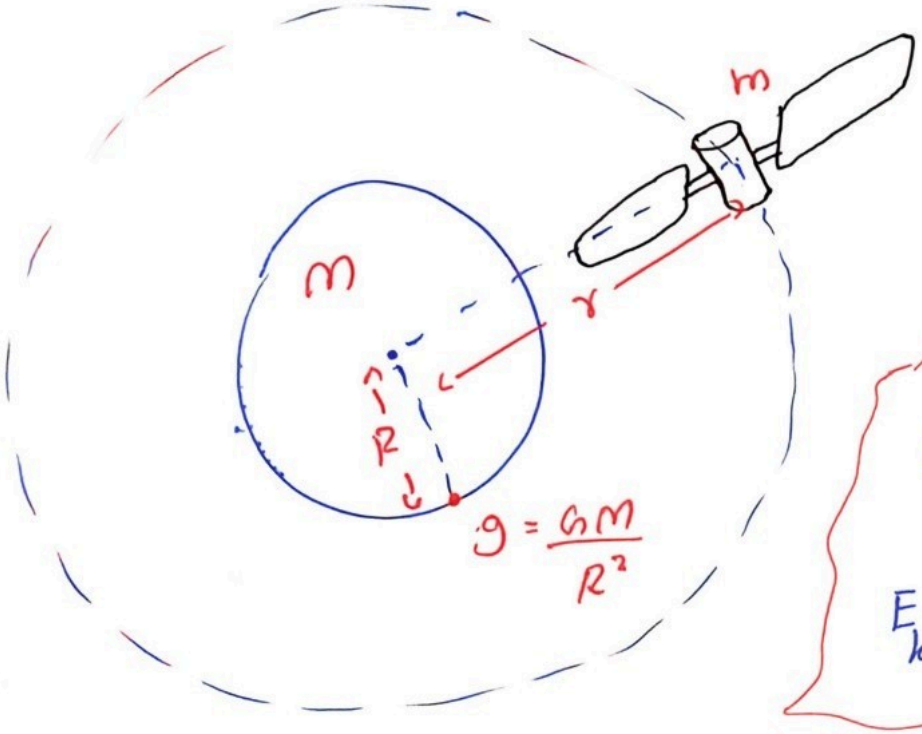


$$E_p = Vm$$

$$E_p = -\frac{GMm}{r}$$

$$E_p = -\frac{gR^2m}{r}$$

(2)



$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$\Downarrow$$

$$E_k = \frac{GMm}{2r}$$

$$E_p = -\frac{GMm}{r}$$

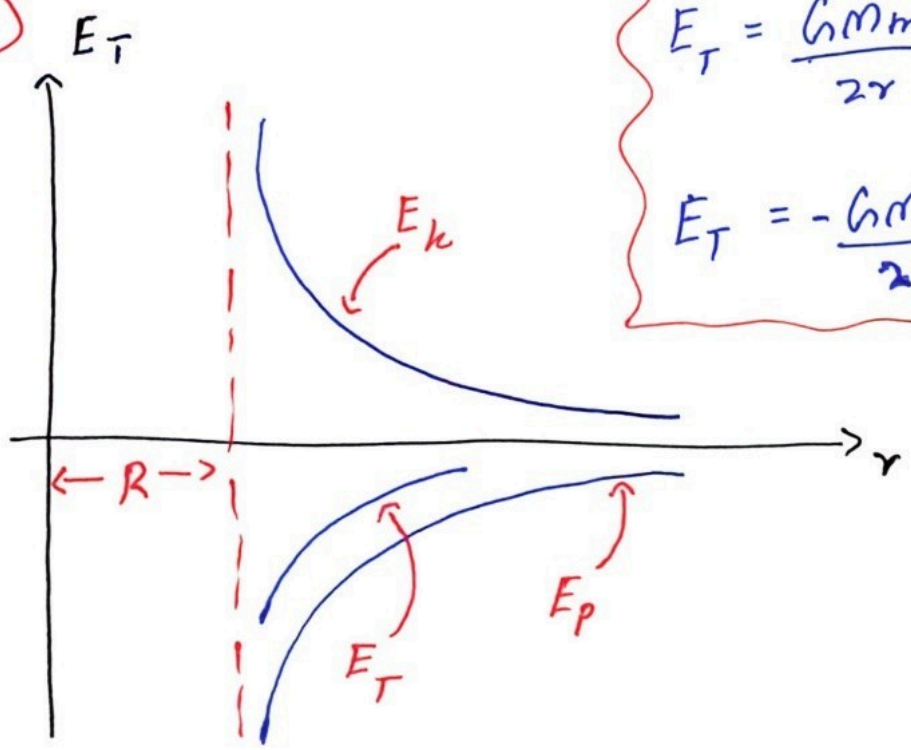
$$E_T = E_k + E_p$$

$$E_T = \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right)$$

$$\Downarrow$$

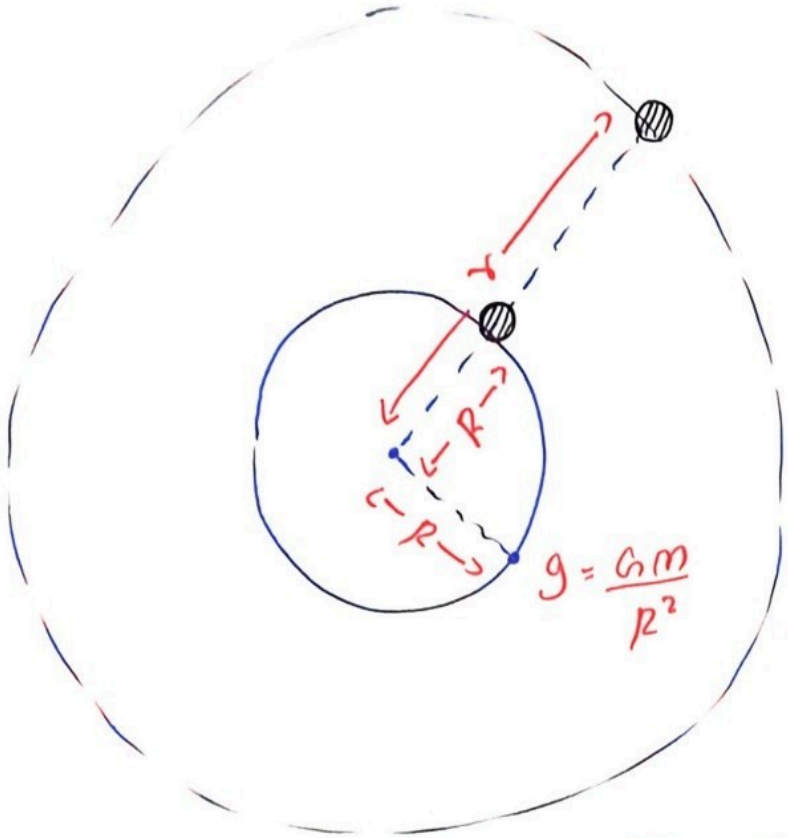
$$E_T = -\frac{GMm}{2r}$$

(3)



AIScanner

(4) The minimum energy need to lift the stellite to circle,



$$E = G.P.E \text{ at circle} - G.P.E \text{ at earth}$$

$$E = -\frac{GMm}{r} - \left(-\frac{GMm}{R}\right)$$

$$E = GMm\left(\frac{1}{R} - \frac{1}{r}\right)$$

$$gR^2m\left(\frac{r-R}{Rr}\right)$$

$$mgR\left(\frac{r-R}{r}\right)$$

AIScanner

→ Total energy,

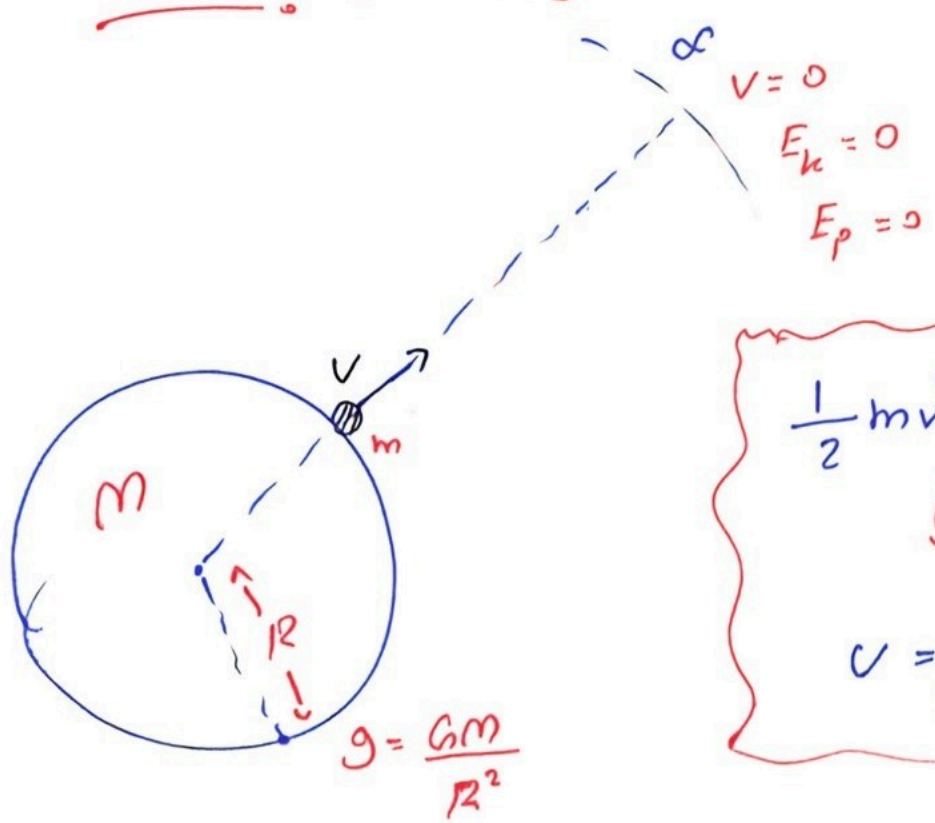
$$E_T = mgR\left(\frac{r-R}{r}\right) + \frac{GMm}{2r}$$

OR

E_T = GPE at circle − GPE at earth + KE need

$$E_T = mgR\left(\frac{2r-R}{2r}\right)$$

→ Escape Velocity



$$\frac{1}{2}mv^2 = \frac{GMm}{R}$$

$$\Downarrow$$

$$v = \sqrt{2gR}$$

AIScanner